HIGHWALL MINING

As semi-underground and semi-surface mining methods, the auger and highwall mining methods are used to extract the remaining coal reserve after the economic threshold imposed by the stripping ratio for surface mining operations is reached. Long circular auger holes or rectangular entries are punched into the coal seam from the highwall left by the previous surfacemining operations to recover part of the remaining coal reserve. Small pillars are left to separate the holes or entries and to prevent the overburden strata from collapsing. Figure 1 shows a working site and some of the potential safety problems of a highwall mining operation.

Figure 1 A Highwall Mining Operation Site (Coal Country)

Due to the recent technological advancements to achieve a much larger penetration length of up to 400 m, the highwall mining method is preferred over auger mining. It continues to grow as an important coal production method in India and the US.

Highwall Mining Method

The highwall mining method is a relatively new semi-surface and semi-underground coal mining method that evolved from auger mining. In a highwall mining operation, the coal seam is penetrated by a **remotely controlled continuous miner (CM)** that is propelled by a **hydraulic push-beam system** that doubles as the coal conveyor, as shown in Figure 2.

A typical cutting cycle includes:

1. sumping (forward pushing) and shearing (cutting drum to cut up and down to extract the entire height of the coal seam).
2. As the coal recovery cycle continues, the CM unit is progressively pushed into the coal seam for about a 6 m increment before another push-beam is inserted.
3. The push-beam system can allow the mining system to penetrate nearly 300-400 m into the coal seam from the mouth.

Currently, the highwall mining systems are capable of handling coal seam thickness as thick from 0.75 to 5 m with a dip up to 8 degrees.

Remote detection systems, such as video imaging, gamma ray sensor and geo-radar could be mounted on the CM unit for the following two purposes:

(1) locating roof and floor rocks to avoid cutting into such rock, and
(2) determining the pillar width left between the previous and current entries to maintain the pillar design width.

Once a production entry, between 9.5 and 11.5 ft wide, is completed, the machine will move to another predetermined location to start another production entry. A continuous solid coal pillar with a pre-determined width will be left between the two adjacent production entries. Since roof bolting, ventilation and machine tramming are not required and the entire production cycle is a continuous one, highwall mining can be very productive. A highwall miner can produce thousands of tons of coal from a bench previously left by contour mining operations, surface mine pits, etc. A mining crew requires 3 to 7 miners. A highwall miner typically produces about 120,000 tons per month.

Figure 2 Schematic of Highwall Mining (**www.prweb.com/releases/2008/06/prweb992884.htm**)